Future-Proofed Workplaces - Smart Connected Real-Estate Integration Demo

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Introduction

About this Demo¶

The described demo environment effectively showcases Cisco's Future Proofed Workplaces vision for Smart Connected Real Estate by integrating networking, collaboration, IoT sensing, data analytics, and automation to create intelligent, flexible, and sustainable workspaces that adapt to evolving business and employee needs.

Key features of this demo environment include:

- **Flexible Deployment:** The demo can be configured with a minimal hardware set and managed remotely via WebRDP or VPN, supporting customer or partner-facing demonstrations. Where physical hardware transport is not feasible, software demo capabilities enable presentations without local devices.
- **dCloud Management:** The environment is managed from a Windows jump host, simplifying configuration through WebRDP without requiring VPN backhaul.
- **Integrated Software Components:**
    - A **Splunk** instance demonstrates smart building dashboarding capabilities using telemetry from Cisco and UPoE alliance partner-supported environments.
    - An **Ubuntu Linux** host runs an **MQTT** broker and node-red instance to support default configurations and allow personal customizations.
    - A Windows Active Directory server is included but currently not leveraged.

    - MHT Technologies, a PoE alliance partner, provides management and configuration software supporting lighting, window coverings, smart desks, and other technologies through the **Aida** software interface.
- **Extensibility:** The setup supports adding new third-party devices by installing their software on existing nodes within the environment, offering high flexibility for customization.

This demo environment aligns with Cisco's approach to future-proof workplaces by combining secure, converged network infrastructure with intelligent building automation and analytics, enabling real-time insights, energy optimization, enhanced occupant experience, and seamless collaboration. It leverages Cisco's UPoE+ switches, Meraki sensors and cameras, Splunk analytics, MQTT messaging, and PoE lighting and control systems from partners like MHT Technologies to deliver a comprehensive smart building solution that can be demonstrated physically or virtually with remote management capabilities.

Key Components and Integrations:

- **Cisco UPoE Switches** (Catalyst 9200CX/9300/9400 Series): Provide 60+W Power over Ethernet to power IoT devices such as PoE lighting, sensors, and collaboration endpoints. These switches enable a converged IT/OT network that simplifies management and reduces electrical distribution equipment by supporting PoE-powered lighting, shades, blinds, sensors, and more.
- **Meraki MV Cameras**: Deliver intelligent video surveillance with embedded analytics for occupancy and density monitoring, enabling space utilization insights and security.
- **Meraki MT Sensors**: Monitor environmental conditions including temperature, humidity, air quality (CO2, VOCs), and noise levels to support health, wellness, and sustainability goals.
- **Splunk and Splunk Edge Hub**: Aggregates and visualizes data from various sensors and devices, providing dashboards for occupancy analytics, environmental monitoring, and energy usage, enabling data-driven decision-making.
- **MHT Aida & PoE Smart Lighting**: PoE-powered lighting fixtures integrated with the network to enable energy-efficient lighting control, daylight harvesting, and occupancy-based lighting adjustments.
- **Cisco Webex Control Hub**: Manages and monitors collaboration devices and integrates telemetry data from Webex endpoints, enabling seamless control and analytics of meeting spaces.

- **Cisco Spaces**: Acts as the digital real estate platform that consolidates telemetry and network data from wired, wireless, and IoT devices. It provides granular occupancy data, space utilization analytics, wayfinding, and employee experience enhancements such as room booking and real-time environmental condition displays.

dCloud supported hardware

The demo requires integration with a dCloud gateway device that supports locally connected Cisco and third-party PoE Alliance partner IoT devices and sensors. PoE-capable gateway models are strongly recommended to simplify setup. The currently supported gateways list is available on the dCloud website at the following link. https://content.ciscodCloud.com/content/cisco-dCloud/us/en/home/support/help/using-dCloud/routers-supported.html

Essential Cisco demo hardware

A minimum demo setup should include a Bluetooth Low Energy (BLE) capable, cloud-managed Cisco Wireless Access Point or Meraki MV Security Camera, along with a Meraki MT Sensor. These devices establish the foundational topology for the physical demo environment and enable immediate testing of connectivity and configurations upon connection to a Power over Ethernet Plus (PoE+) port on a supported dCloud gateway. All hardware components must be managed through a cloud management instance, such as the Meraki dashboard, to ensure centralized control and monitoring. An always-on demo option may be available to support continuous operation.

This configuration facilitates rapid deployment and validation of smart building technologies within the demo environment.

Getting started

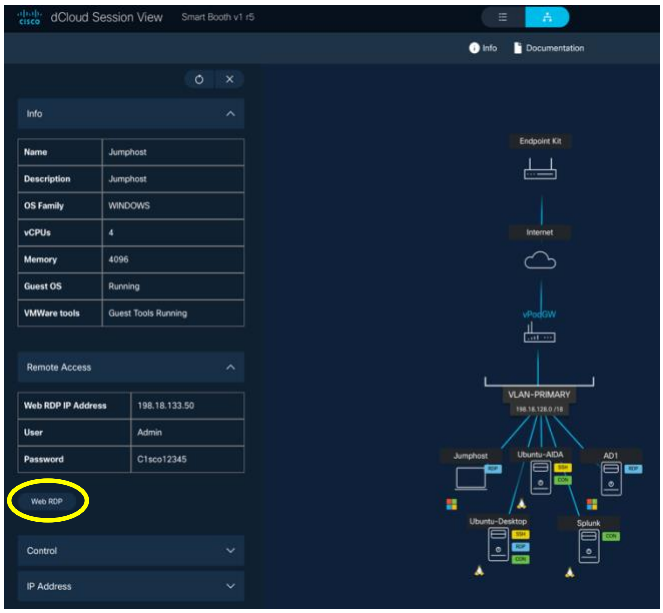
Jumphost

Once the dCloud instance is scheduled and running, verify basic connectivity by following these steps:

1. From the dCloud session view, select the information icon labeled "(!) Info" and then choose "Jumphost." The Jumphost interface should open in a new browser tab.

2. Click the "Web RDP" link (highlighted or circled in yellow) to launch the Jumphost interface within a web browser. If a "Cookies too big/long" error message appears, open the link using an incognito or private browsing window.

3. On the Jumphost desktop, launch the MQTT Explorer application.



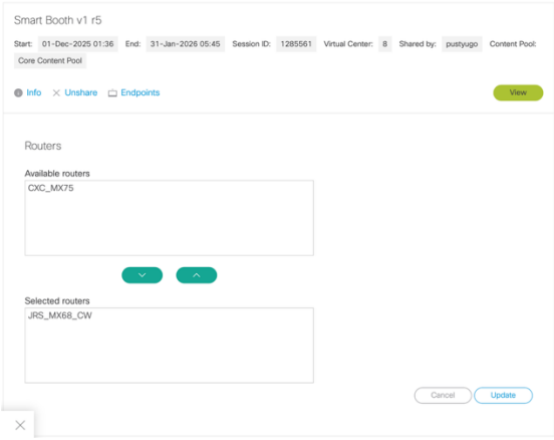
4. Connect MQTT Explorer to the preconfigured Ubuntu MQTT Broker instance saved within the application.

5. Keep MQTT Explorer open to monitor and validate device communications with the MQTT broker during subsequent lab setup or configuration activities.

This process ensures that the dCloud environment and MQTT broker connectivity are operational and ready for further demonstration or testing.

Router

All included links in this demo assume operation from the RTP data center. When using the example links provided, substitute the appropriate data center URL as needed to match the actual environment.



Supported routers can be added to the demo through the dCloud portal at https://dcloud2-rtp.cisco.com/dashboard/routers. If no suitable router is listed, use the "Add New Router" button on the dashboard to provision the demo gateway. Locate the router's serial number on the device box or the bottom of the unit and add it to the list of available routers.

Once the router is added, navigate to the running sessions view at https://dcloud2-rtp.cisco.com/dashboard/sessions. Select the newly available router and update the session. Note that this process may take some time to complete.

VPN Access

All links in this document can be followed directly if connected to the dCloud gateway, or Cisco Secure Client VPN using the instantiated dCloud credentials.

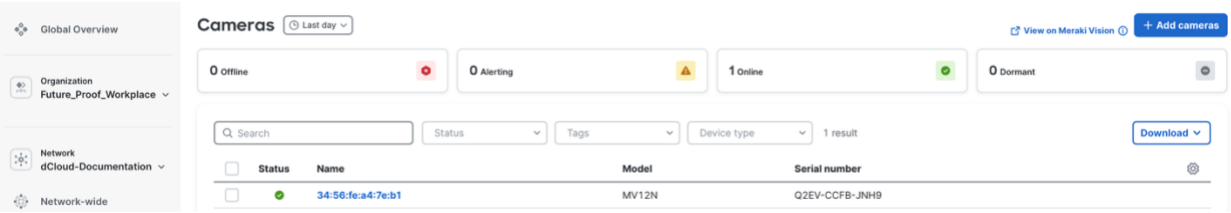
IoT devices

[Meraki MV security camera | Cisco wireless access point]

This demo assumes basic working knowledge of the Meraki Cloud management interface and only covers the configuration of components required to successfully use this demo.

Meraki MV camera

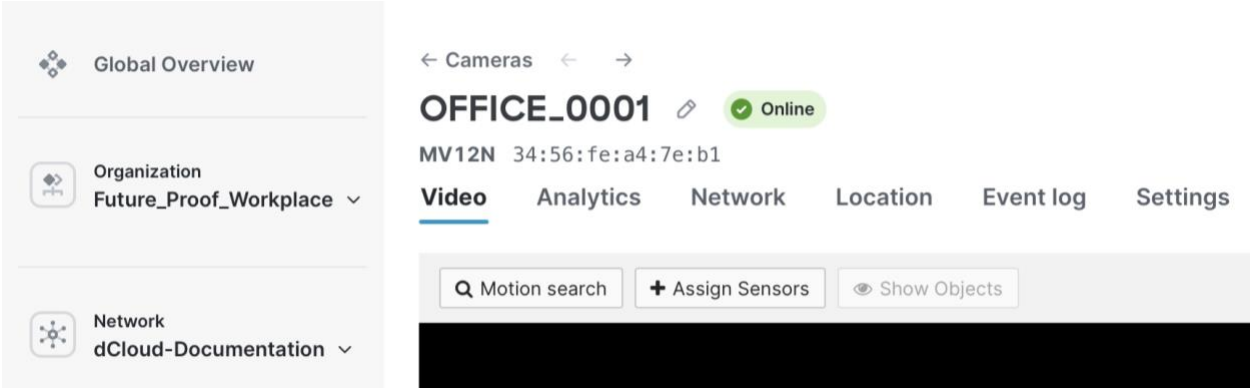
No special configuration actions are required for the camera(s) to perform as an IoT gateway for Meraki MT sensors provisioned in the same Organization and Network. Connect your camera directly to a PoE+ port on the endpoint router and verify it is operational in the Meraki Dashboard.



Device naming

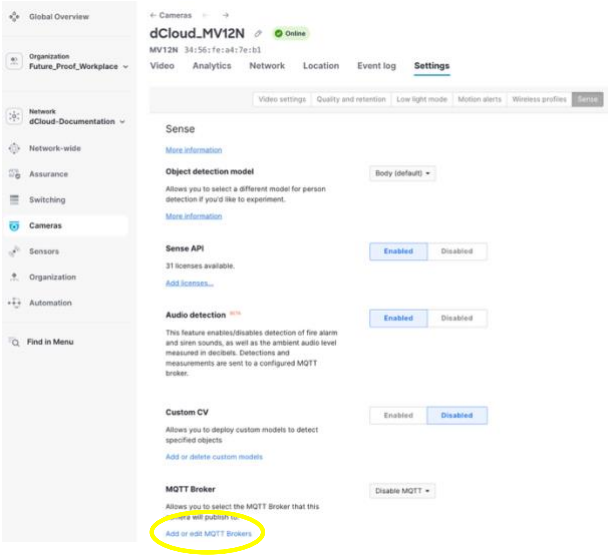
Camera names are used in preconfigured demos / dashboards. To simplify implementation, name cameras based on a room index with an underscore as a delimiter.

Example – OFFICE_0001


Global Overview
Organization
Future_Proof_Workplace
Network
dCloud-Documentation
← Cameras
OFFICE_0001
Online
MV12N 34:56:fe:a4:7e:b1
Video
Analytics
Network
Location
Event log
Settings
Motion search
Assign Sensors
Show Objects

MQTT configuration

Do demonstrate using the Meraki MV camera(s) as a sensor, they must be configured to publish to an MQTT broker.

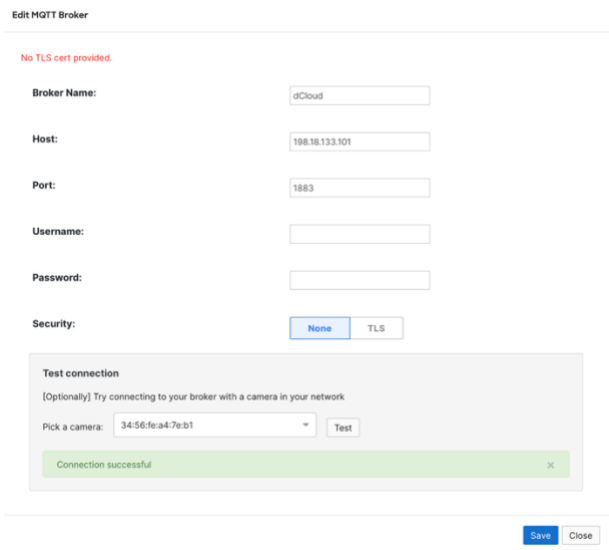


In the individual camera configuration, under **Settings** and the **Sense** tab there are a number of items to configure.

Confirm **Object detection model** is *Body (default)*
Set the **Sense API** to *Enabled*
Set the **Audio detection** to *Enabled*
Leave **Custom CV** set to *Disabled*

MQTT Broker

Click **Add or Edit MQTT Brokers**
Set **Broker Name:** to *dCloud*
Set **Host:** to *198.18.133.101*
Set **Port:** to *1883*
Ignore **Username:**
Ignore **Password:**
Confirm **Security:** set to *None*

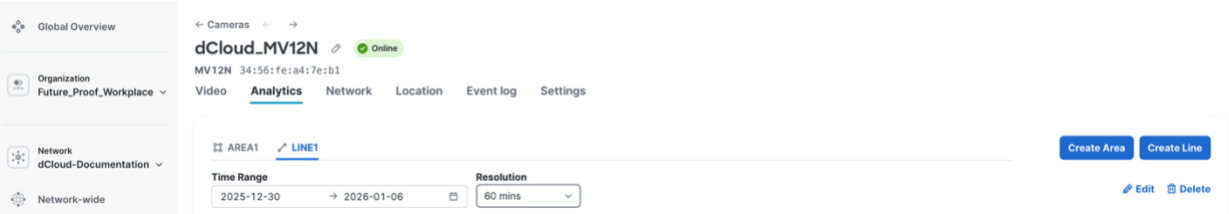


It will save time later to test this configuration now. Pick one of the connected cameras and test the connection.

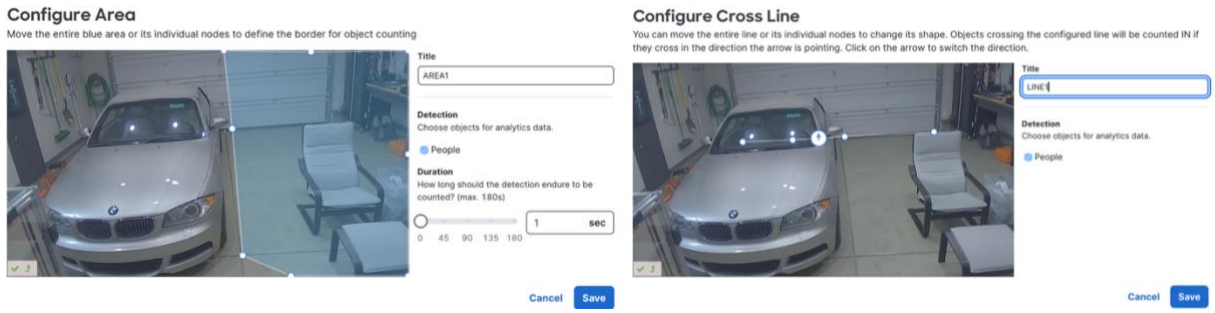
Save the MQTT Broker configuration.
Select your newly configured MQTT server and save your camera settings in the cloud dashboard.

Occupancy and traffic

If camera(s) are placed where in demo locations, configure occupancy area(s) and traffic lines. Select the Analytics configuration and create an area and a line to configure these features. Please note, analytics features may vary depending on camera generation and sense licensing.



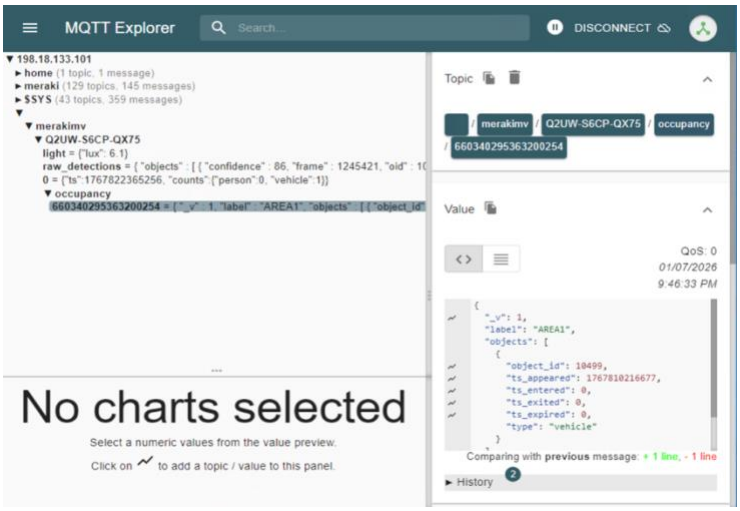
Demo Splunk Dashboards and demo Node-RED workflow / automations are pre-configured to respond and report on MQTT occupancy and traffic topics containing in **AREA_0001** and **LINE_0001** labels. If this demo instance does not require customization, using these names/labels will simplify demo setup.



With more than one camera it is still possible to isolate events by MV serial number…

Verify Configuration

Return to the Jumphost and validate in MQTT explorer that messages are being published by the camera(s). Expand each topic and subtopic by clicking the triangle icons. If the configuration is present for areas / line crossings, there will be occupancy and traffic topics published.



Cisco wireless access point

While MQTT configurations are not required for Meraki MT Sensors to use access points as a BLE IoT Gateway the telemetry can provide interesting data later from within the Splunk dashboard.

IoT Radio Settings

From the **Wireless** menu select the **IoT Radio Settings** under **Configure**.

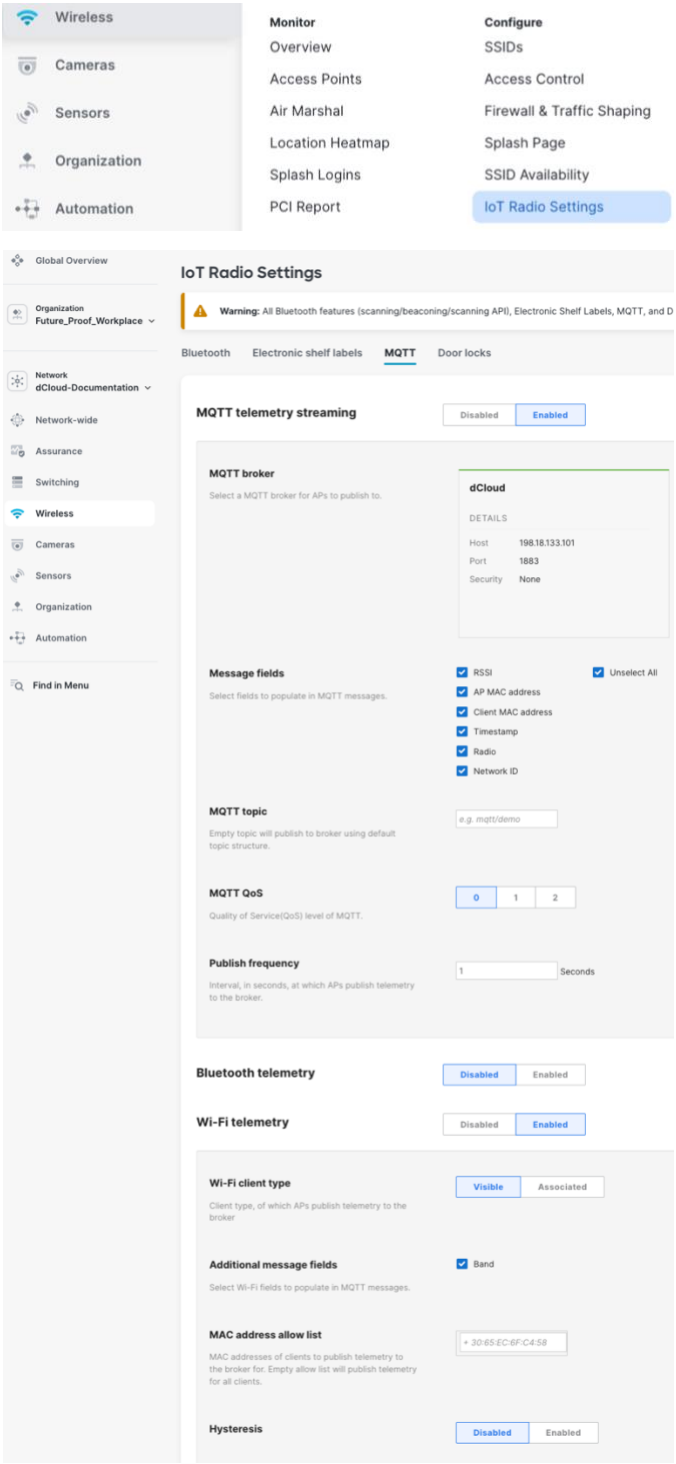
This will open configuration options for this network. Select the MQTT tab and configure your environment as follows:

MQTT telemetry streaming to *Enabled*
MQTT Broker

Click **Add or Edit MQTT Brokers**
Set **Broker Name:** to *dCloud*
Set **Host:** to *198.18.133.101*
Set **Port:** to *1883*
Ignore **Username:**
Ignore **Password:**
Confirm **Security:** set to *None*

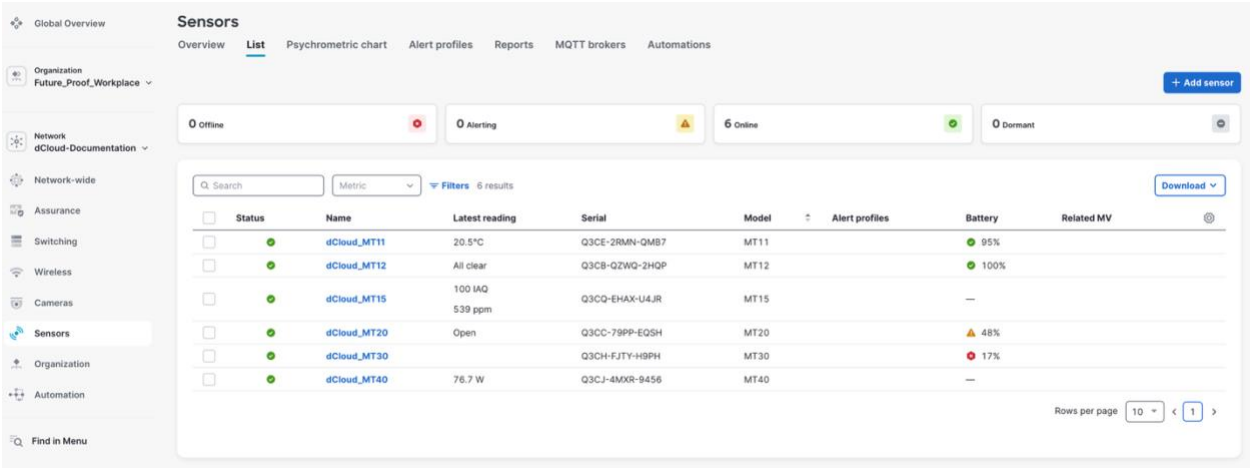
Messaging fields leave all enabled
MQTT Topic Leave *default*
MQTT QoS set to *0*
Publish frequency set to *1* Seconds
Wi-fi telemetry set to *Enabled*
Wi-fi client type set to *Visible*
Additional message fields select *Band*
MAC address show list leave *empty*
Hysteresis set to *Disabled*

Once the configuration is saved, verify messages are present in MQTT Explorer on the Jumphost.

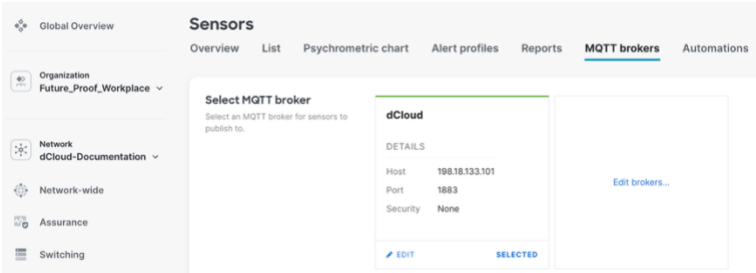


Meraki MT sensors

Once MT Sensors are added to the same network as the MV cameras or wireless access points, they will begin to report telemetry to the Meraki Dashboard.



Once the sensor status is online, proceed to configuring the MQTT broker for sensors in this network. Unlike the wireless IoT configuration, the MT Sensors can reuse the profiles created in the camera configuration. Edit the brokers and select the previously created dCloud broker.



Verify basic configuration

Jumphost

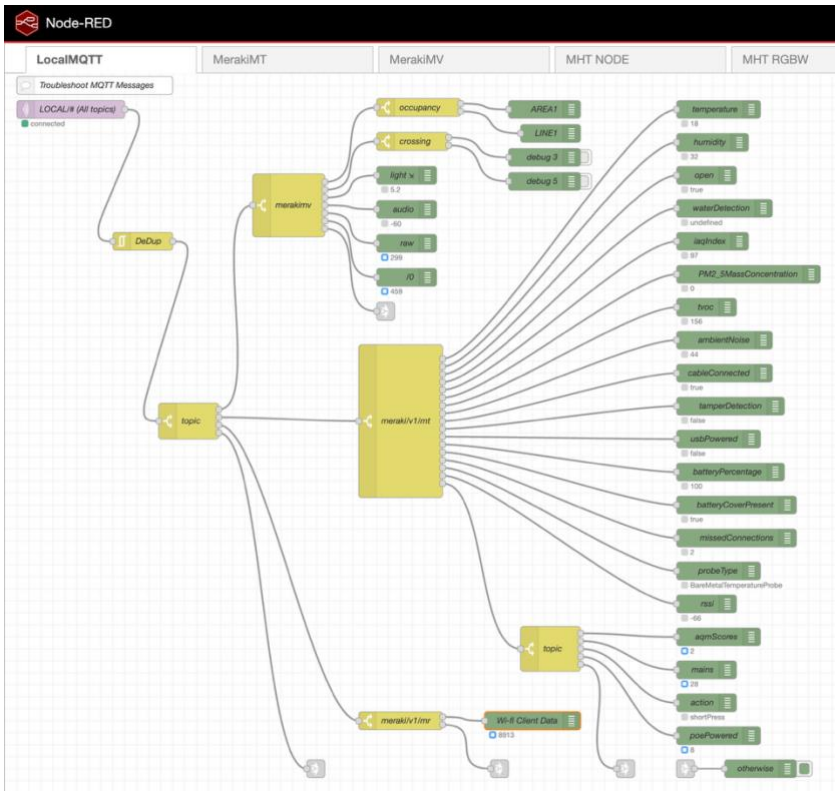
Return to the Jumphost and verify that MQTT Explorer sees messages published under the following topics:

/merakimv/# (displays telemetry from the MV cameras)
meraki/v1/mt/# (displays telemetry from the MT sensors)
meraki/v1/mr/# (displays telemetry from the Wi-fi access points)

From the Jumphost access is provided to the web interfaces of several other servers and services. Open chrome to use these services to verify the configuration now.

Node-RED

Node-RED is provided as a **flow-based, low-code development tool** designed for visual programming. It is primarily aimed at wiring together hardware devices, APIs, and online services, especially in the context of the Internet of Things (IoT).



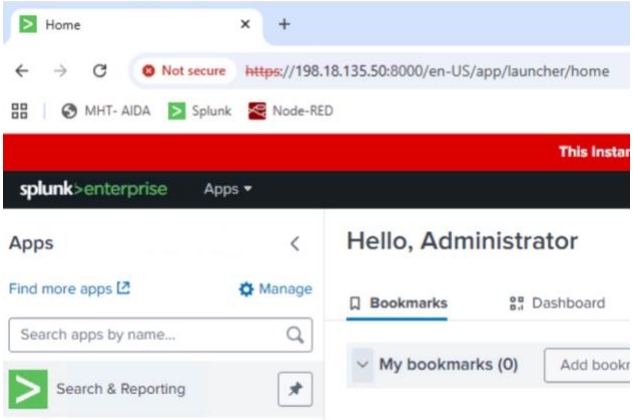
It is provided here as a mechanism to interconnect the devices and services present within this demo, and to facilitate automation and orchestration of activities and actions. Like MQTT explorer the **LocalMQTT** flow (illustrated as a tab below) is subscribed to all topics published on the local MQTT instance. The yellow switch nodes look at MQTT topics and payload object data and forward it to the green debug nodes which display the last message received or a message count based on filters.

The otherwise node in the bottom right is a catchall. This **LocalMQTT** flow is provided as a sample flow to view messages, it does not perform any orchestration or automation activities.

Flows and nodes can be easily exported if saving customizations is required. Exporting nodes is highly recommended if customizations are performed to protect / save work. You can also export to, to recreate this flow in a local Node-RED application / Splunk Edge Hub container.

Splunk

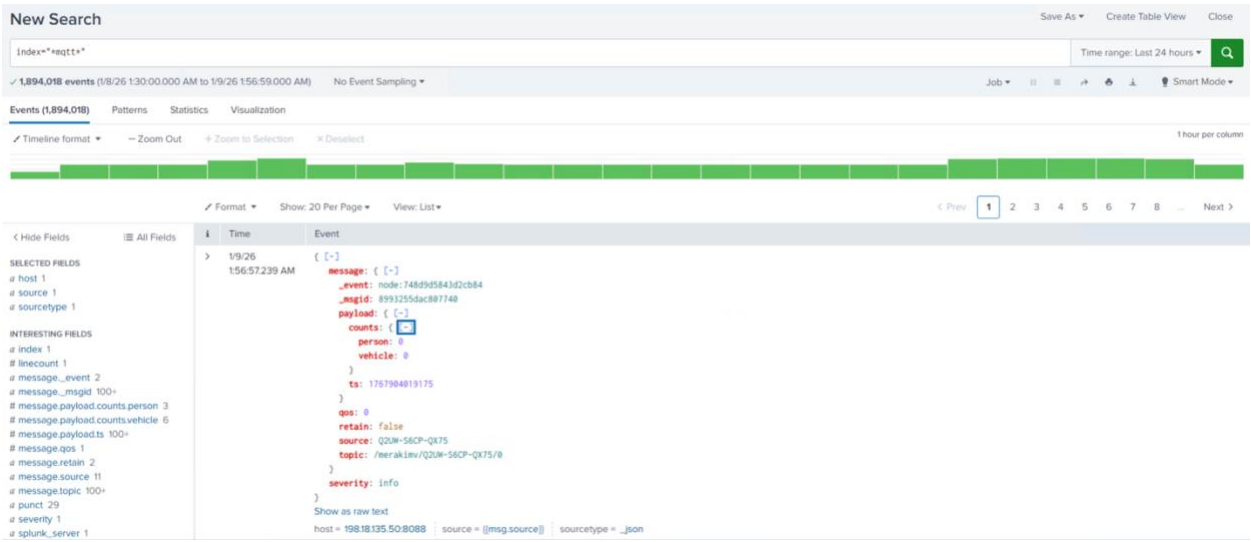
Validating that the data is being ingested into the dCloud Splunk instance is easy, Node-RED is configured to forward messages from the Ubuntu MQTT Broker to the Splunk server running at https://198.18.135.50:8000



Validation: Use the Jumphost to connect to the Splunk instance and click on **Search & Reporting**

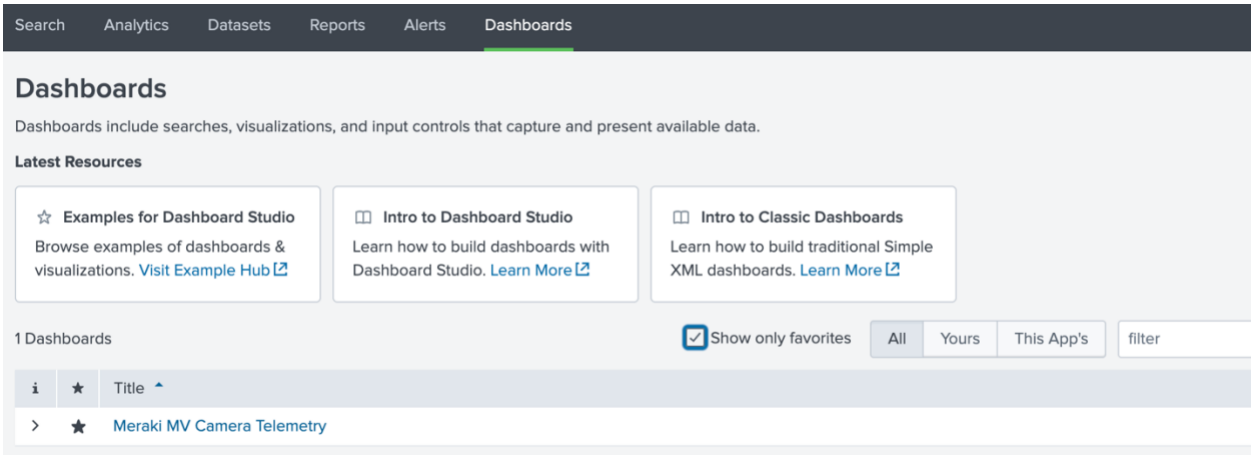
In the **Search & Reporting** application enter *index="*mqtt**"* and press enter to search the default time range of the last 24 hours for inbound messages to the Splunk environment.

Select one of the results and use the + signs to expand / view the data returned by the search. If everything is working properly you should be seeing data from [cameras | access points] and sensors in the demo environment.

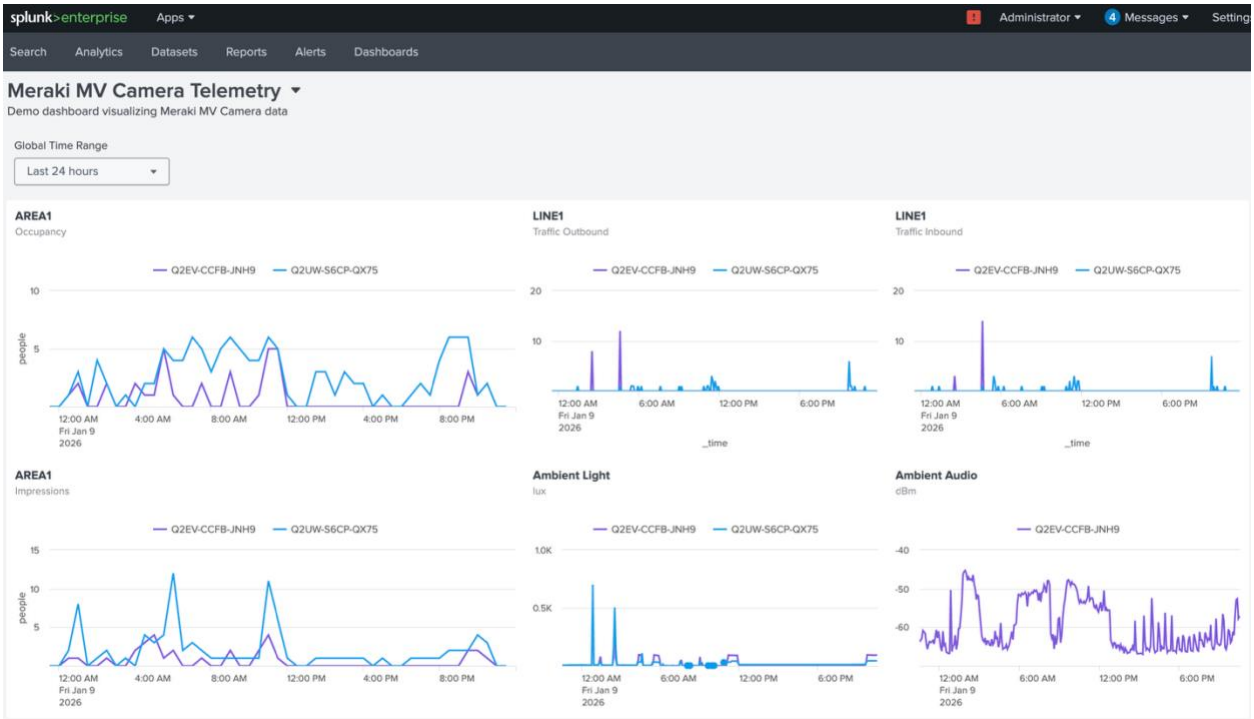


Minimal Demo

There are some previously saved dashboards you can use out-of-the-box. In the Search application open the dashboards tabs and click the button to show favorites only.



For example: opening the Meraki MV Camera Telemetry dashboard https://198.18.135.50:8000/en-US/app/search/meraki_mv_camera_telemetry should display the following with the configured camera telemetry data.



It should be noted that some camera models can have their microphones "physically" disconnected by using a jumper/switch on the back of the device. Devices configured with this feature will never measure or react to audio.

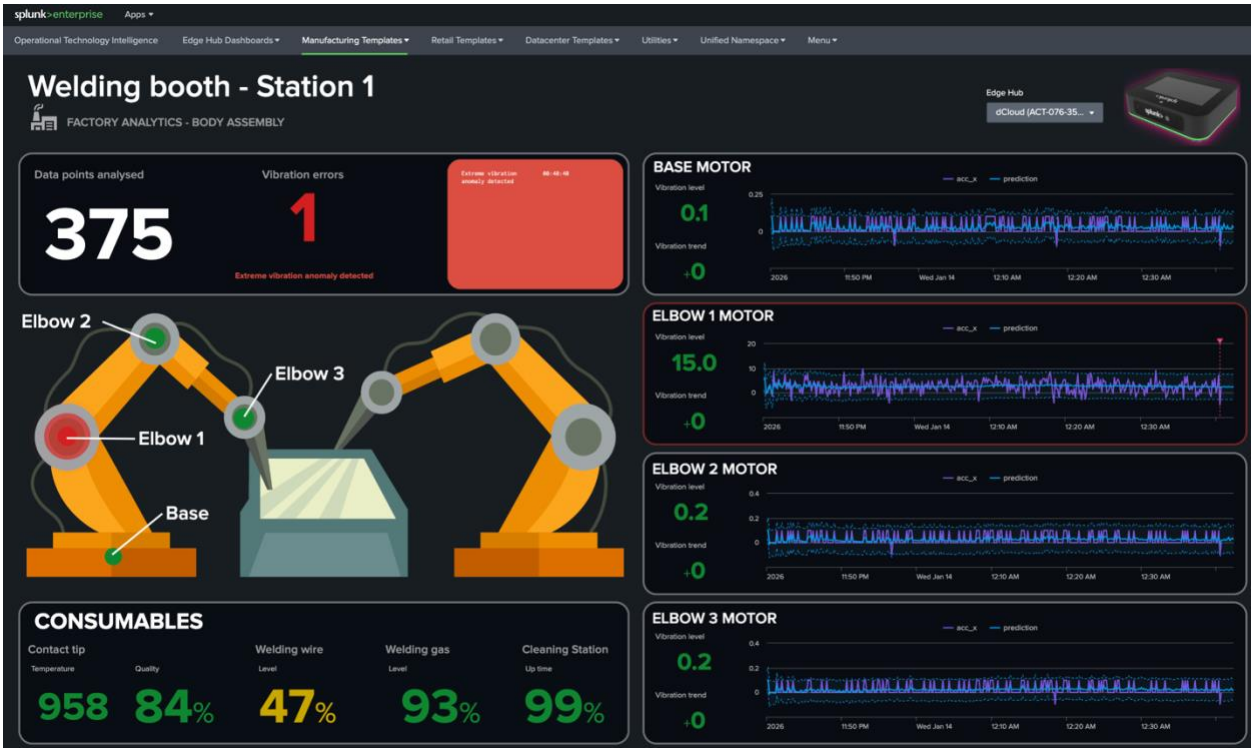
Expanding the demo

Splunk Edge Hub

This demo instruction cannot cover all the setup requirements for Splunk Edge Hub (SEH), please follow the directions at (THIS LINK) to add a Splunk Edge Hub to the Splunk server running at https://198.18.135.50:8000

Once your SEH is registered to the dCloud Splunk instance, it is relatively simple to demonstrate basic IoT dashboards.

Open https://198.18.135.50:8000/en-US/app/splunk-app-ar/manufacturing_vibration_analytics and select the configured SEH in the Edge Hub dropdown which will display a dashboard similar to the one seen below. There are several other sample dashboards here which can be used to tell the FPW IoT integration story.



Lighting things up

Basic Hardware

- *1x MHT Node 90*
- 1x MHTi-DL-SC downlight (alternates can be substituted however it is important to understand that actual configuration requires specialized knowledge, and may not result in desired outcomes or success
- 1x MHTi-WS

Datasheets

https://mht-technologies.com/wp-content/uploads/2025/12/MHTi-NODE-90-Consolidated-V7.pdf

https://mht-technologies.com/wp-content/uploads/2024/05/MHT-DL-Series.pdf

https://mht-technologies.com/wp-content/uploads/2024/02/MHTi-WS-Spec-Sheet.pdf

Connection

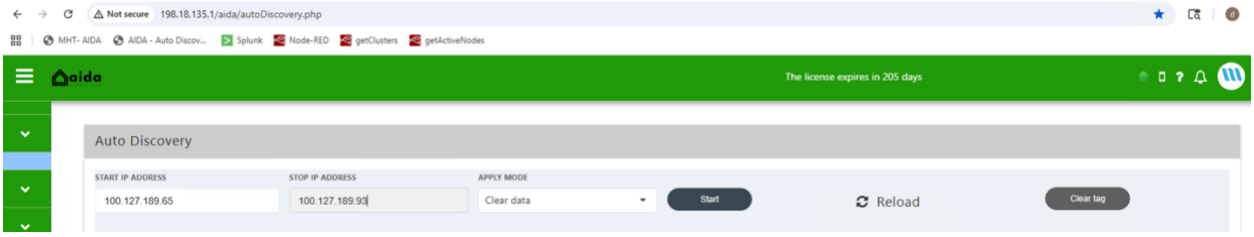
While a single downlight or even possibly 2 downlights can be configured and supported from PoE+ ports on the dCloud Gateway or an MS-130, Any substantial demonstration will require the capabilities of a UPoE or UPoE+ capable switch. Consideration should be given to the Catalyst C9200CX-8UXG-2X running in device managed mode in the Meraki Dashboard.

One of the key values of the physical demo is the ability to demonstrate how easy it is to and control PoE lighting. Using standard category cabling, interconnect each component as follows. It is important to understand that the purple category cables shown on the right are being used as “power” connections when connected to the downlights on the CH1 and CH2 ports, or RS485 data connections when connected to the Console port.

Only the green cable on the PoE port provides uplink power and ethernet transport.

Configuration

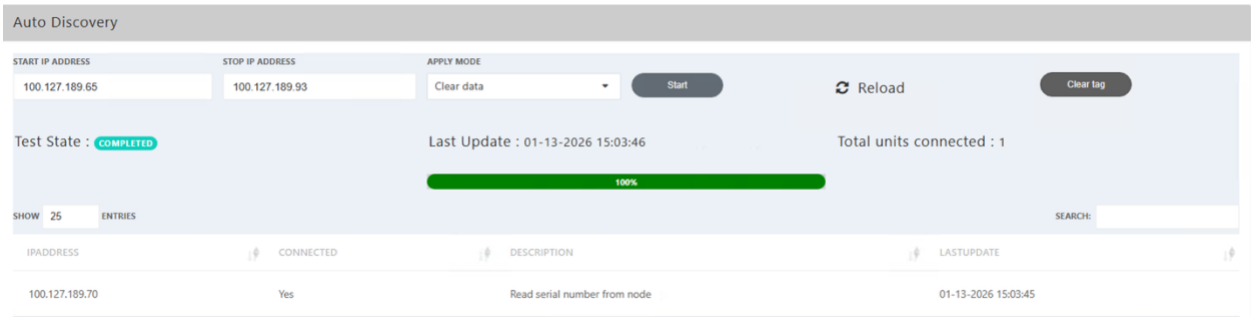
On the Jumphost or a computer connected to the dCloud environment, open the AIDA software interface and discover the newly connected PoE LED driver. There is an auto-discovery shortcut found in the browser bar which forwards to the following link: http://198.18.135.1/aida/autoDiscovery.php



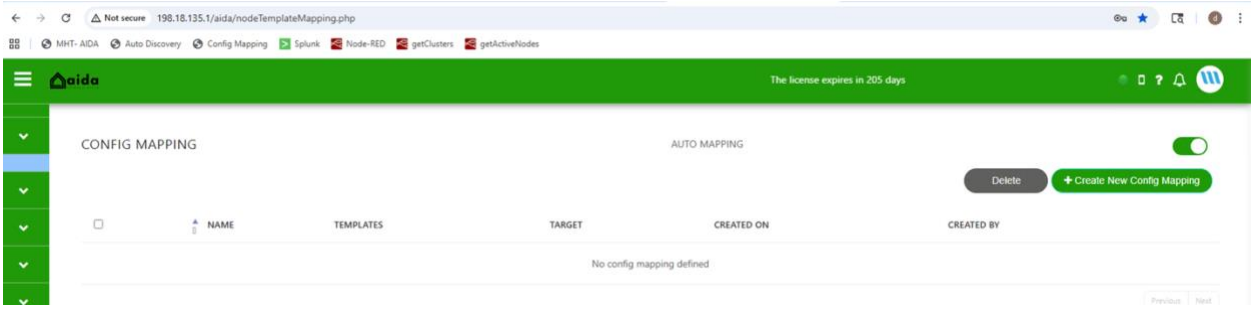
The start IP address and end IP address is determined by the subnet the dCloud Gateway has provided to the network side of the device. To determine what addresses to search, use the "Get Endpoint Kit Info" shortcut on the Jumphost dashboard. Alternately look at one of the connected devices in the Meraki dashboard like a camera to determine its IP Address, or a local computer connected to the dCloud gateway. Currently the dCloud gateway subnets are configured one wired, or two (wired and wireless) /27 networks. Once the network the driver(s) is connected to has been discovered,complete the values in the Auto Discovery window for that subnet as shown above and then clear *Start*

START IP ADDRESS to *100.127.x.FIRST*
END IP ADDRESS to *100.127.x.LAST*
APPLY MODE to *Clear data*

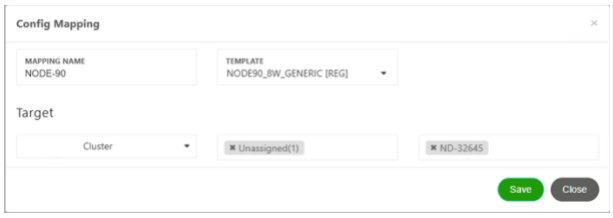
The discovery script will run for a short time and should find the connected driver(s).



Once discovered, proceed to configuration mapping by following the link in the bookmarks bar on the Jumphost or opening http://198.18.135.1/aida/nodeTemplateMapping.php on a computer connected to the dCloud gateway.



Click **+Create New Config Mapping** and set the values in the following configuration / dialog box and click **Save**

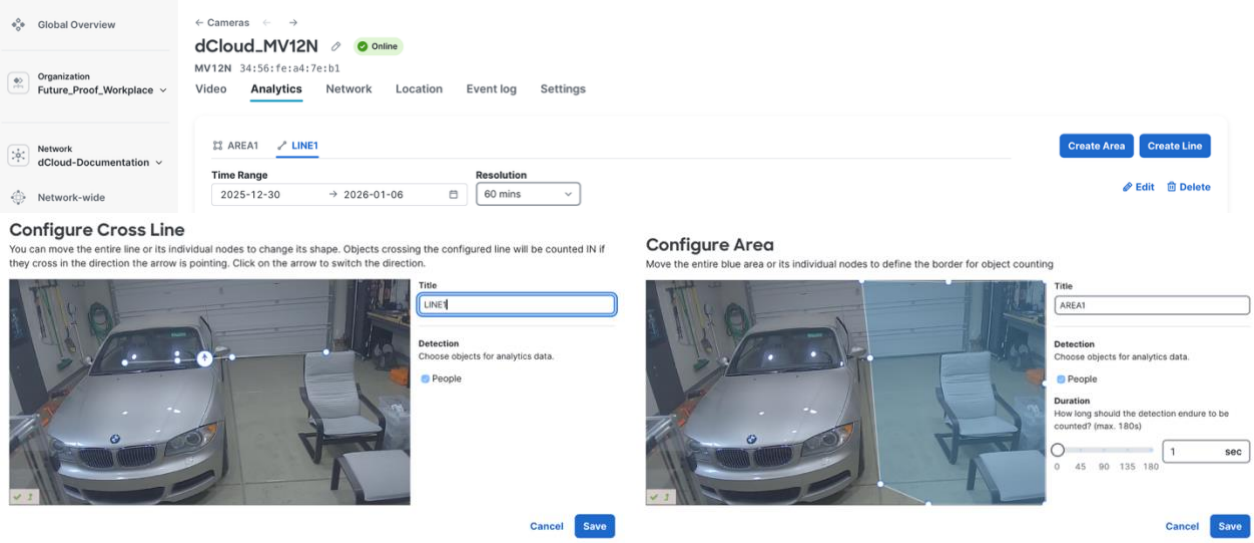


MAPPING NAME to *NODE-90*
TEMPLATE to *NODE90_8W_GENERIC [REG]*
TARGET to *Cluster*, *Unselect All*, *Select Unassigned*, *Unselect All*, *Select ND-#####*

at however it has been “pre-configured” to make integration quick and easy. ecommendation So you've decided that you want to run the future proof workplace dCloud demo to demonstrate integrations in smart building solutions that's great but to get started you're going to need a minimum of a dCloud supported endpoint loosely translated as an MX gateway a sensor of some kind from meraki and a camera or access point that is capable of seeing the Bluetooth over the air messages from that network so let's start with step one minimal configuration of the network connect your MX gateway MT15 or other sensors and your Merc amra that's BLE capable or your meraki access point that's BLE capable two your dCloud configured gateway or endpoint use the Poe ports on your device to add the empty 15 and the camera or used other battery operated sensors in combination with the MT15 a little side note if you need to use the other po E port for something else you can connect the MT15 to the USB port on the side of the MX it doesn't need a lot of power but this sort of loses the impact of the I'm Poe capability that we'll get to a little bit later in the demo so once you have those connected you need to go to your meraki environment and provision the devices or search for the devices inside of your meraki environment so

If your camera is currently placed where your demo will be performed, you may want to also configure your occupancy area(s) and traffic lines at this time. Do this by selecting the analytics tab for your camera. Click create area or create line to configure these features. Please note, analytics features will vary depending on your camera generation and licensing.

Demo Splunk and Node Red automations are configured to look for MQTT occupancy event



topics ending in **AREA1** and **LINE1** by default. If your demo does not require customization, using these names will simplify your demo setup. If you have more than one camera you will be able to isolate each topic by MV serial number…

You can verify that your cameras are generating information by walking through the lines or lingering in the areas and seeing if MQTT events are created. The easiest way to do this is by logging into the dCloud nodeRed instance at http://198.18.133.101:1880.